

Comparing Schedule-Based Low-Rank Functionals with KSS, BLM, and Borovičková-Shimer

Scope and source convention

This note reports a seven-design, 100-replication Monte Carlo comparison of the project's schedule-based estimands with additive fixed-effects/KSS, grouped BLM, and Borovičková-Shimer procedures. The simulated complete wage schedule makes every population target observable, so estimation error and estimand differences can be evaluated separately.

The project procedure implemented here is the **low-rank plug-in without LOO correction**. The project's full leave-worker-out estimator is still under development and is not simulated.

Statements marked **Source** reproduce a definition, method, design choice, or numerical output from the cited paper or repository file. Statements marked **Derived** follow algebraically from those definitions or summarize the reported Monte Carlo output. No literature result is attributed to the simulation, and no simulation result is attributed to the cited papers.

Source: SIMULATION_CONTRACT.md; configs/full_ladder.json; PRODUCTION_RESULTS.md; reporting commit 8922da6; production configuration fingerprint e03e0c242706502fcd58f20c1cc52c4deeb234f48c981a4885e13157d6ac2a09.

Notation

Symbol	Meaning	Provenance
i, j	Worker and firm indices; I, J are their population counts.	Source: project simulation contract.
Y_{ij}	Observed wage or log wage for worker i at firm j .	Source: project model.
$X_{ij}' \delta$	Observed covariate component. It is set to zero in the current simulation.	Source: project model and simulation contract.
α_i, ψ_j	Additive worker and firm components.	Source: project model.
u_i, v_j, Λ	Worker factor, firm factor, and interaction matrix; $\text{rank}(\Lambda)=r$.	Source: project model.
$M=(m_{ij})$	Complete systematic wage schedule generated in each replication.	Source: simulation contract.
P_{obs}	Population assignment law over worker-firm pairs.	Source: simulation contract.
p_i, q_j	Worker and firm marginals of P_{obs} .	Source: simulation contract.
μ, a_i, b_j, h_{ij}	Product-weighted grand mean, worker main effect, firm main effect, and interaction in $m_{ij}=\mu+a_i+b_j+h_{ij}$.	Source: simulation contract.
Q_F	Mean across workers of the firm-side variance of their wage schedules.	Source: project estimand definition.
H_F	Mean pairwise firm contrast variance across workers.	Source: project estimand definition.
ρ_H	Normalized interaction share, $H_F/(2 Q_F)$.	Derived from the project definitions.
C_{assign}	Half the difference between systematic-wage variance under P_{obs} and under $p \times q$.	Source: project estimand definition.
$V_{\psi}^{\text{AKM}}, C_{\psi\alpha}^{\text{AKM}}$	Firm-effect variance and worker-firm covariance in the assignment-weighted additive projection.	Derived comparison target defined in the simulation contract.
A_{1k}^{BLM}	Mean wage for latent worker type 1 and firm class k .	Source: BLM grouped model; local stationary specialization.
ρ_{BS}	Correlation between Borovičková-Shimer worker and firm wage types on observed matches.	Source: Borovičková and Shimer (2020); local population analogue.
$r_{\text{hat_BIC}}$	Rank selected by the exploratory common-support BIC comparison.	Source: local implementation; not part of the unfinished LOO theory.
e_b	Replication error, estimate minus its declared population target.	Derived Monte Carlo notation.

1. Model, estimands, and the comparison problem

The project model is

$$Y_{ij} = X_{ij}' \delta + \alpha_i + \psi_j + u_i' \Lambda v_j + \epsilon_{ij}.$$

Each replication generates the full systematic schedule M , an assignment law P_{obs} , and a noisy finite panel. Under the product reference distribution $p \times q$, the schedule has the canonical decomposition

$$m_{ij} = \mu + a_i + b_j + h_{ij}.$$

The four headline project truths are

$$Q_F = \sum_i p_i \text{Var}_q(m_{iJ}),$$

$$H_F = \sum_{jk} q_j q_k \text{Var}_p(m_{Ij} - m_{Ik}),$$

$$\rho_H = H_F / (2 Q_F),$$

$$C_{\text{assign}} = 0.5 \{ \text{Var}_{\text{Pobs}}(m_{IJ}) - \text{Var}_{p \times q}(m_{IJ}) \}.$$

They satisfy $Q_F = \text{Var}_q(b_J) + H_F/2$. Under additive separability, $h_{ij}=0$, so $H_F=\rho_H=0$, $Q_F=\text{Var}_q(b_J)$, and C_{assign} reduces to the assignment covariance between worker and firm main effects.

Source: SIMULATION_CONTRACT.md, Sections 1-2. Derived identity: substitute the product-weighted decomposition into the four definitions and use the zero-mean restrictions on a_i , b_j , and h_{ij} .

The central comparison is not always estimator against estimator for a common target. Under nonadditivity, AKM/KSS estimates moments of the assignment-weighted additive projection of M , whereas the project objects remain functionals of the complete schedule. Consequently,

$$\text{estimate}_p - \theta_{\text{project}} = (\text{estimate}_p - \theta_{p^*}) + (\theta_{p^*} - \theta_{\text{project}}).$$

The first term is estimation error for procedure p ; the second is an estimand or model gap. The simulation reports both.

Derived decomposition from the declared native target θ_{p^} and project target θ_{project} ; see SIMULATION_CONTRACT.md, Section 5.*

2. Procedures

2.1 Low-rank plug-in without LOO correction

For a fixed interaction rank r , the current project procedure minimizes

$$\sum_{(i,j) \in E} n_{ij} [Y_{ij} - \alpha_i - \psi_j - u_i'v_j]^2,$$

using match counts n_{ij} as weights. The implementation uses alternating weighted least squares, spectral plus perturbed initializations, factor centering, and SVD normalization. Positive-rank fits use a connected support core with at least $r+2$ distinct matches per retained worker and firm. The exploratory BIC selector compares candidate ranks on a common support sample. Returned fits can be classified as stable or unstable; unstable values remain in the unconditional procedure-performance calculation.

No quadratic or leave-worker-out correction is applied. This distinction is essential: the procedure is a low-rank completion plug-in benchmark, not the project's proposed full LOO estimator.

Source: SIMULATION_CONTRACT.md, Sections 4 and 7; src/loo_sim/low_rank.py; src/loo_sim/monte_carlo.py.

2.2 AKM and KSS

The additive benchmark estimates worker and firm fixed effects and reports the plug-in firm variance and worker-firm covariance. KSS-HO applies the homoskedastic correction; KSS-HE applies the heteroskedastic leave-out correction. Kline, Saggio, and Sølvesten develop leave-out estimators for quadratic forms in linear models with unrestricted heteroskedasticity.

Under nonadditivity, the simulation evaluates AKM/KSS against its declared native target: the assignment-weighted additive projection of M . It does not assume that a correction derived for the additive linear model recovers the schedule-based project functionals after an omitted interaction is added.

Source: Kline, Saggio, and Sølvesten (2020), Econometrica 88(5), 1859-1898, DOI 10.3982/ECTA16410; local implementation: PyTwoWay 0.3.21 and src/loo_sim/pytwo_way_estimators.py; target definition: SIMULATION_CONTRACT.md, Section 5.1.

2.3 Grouped BLM

BLM models earnings distributions with two-sided worker and firm heterogeneity and permits nonlinear interactions. Its tractable two-step implementation classifies firms before estimating worker-type by firm-class earnings distributions. The simulation uses the stationary grouped mean target A_{1k}^{BLM} and compares oracle firm groups with wage-distribution clusters. Label alignment is used only to evaluate Monte Carlo loss.

Source: Bonhomme, Lamadon, and Manresa (2019), *Econometrica* 87(3), 699-739, DOI 10.3982/ECTA15722; local specialization: SIMULATION_CONTRACT.md, Section 5.2, and src/loo_sim/blm.py.

2.4 Borovičková-Shimer

The Borovičková-Shimer procedure estimates the correlation between worker and firm wage types while allowing many workers and firms to have only a small number of observations. The simulation evaluates it against its native population analogue,

$$\rho_{BS} = \text{Corr_Pobs}(\lambda_{I^{BS}}, \mu_{J^{BS}}),$$

where each worker type is the worker's mean systematic wage over observed matches and each firm type is the corresponding firm mean. This correlation is not relabeled as the project assignment covariance.

Source: Borovičková and Shimer (2020), "High Wage Workers Work for High Wage Firms," February 2020 manuscript; earlier version NBER Working Paper 24074, DOI 10.3386/w24074; local target definition: SIMULATION_CONTRACT.md, Section 3.

3. Monte Carlo design

The ladder changes one economically relevant feature at a time before adding the grouped and rank-misspecification designs.

Design	Population	Assignment/sorting	Panel	Candidate ranks
Additive, independent	80 workers, 10 firms, rank 0	Independent	T=7	0, 1
Additive, common sorting	80 workers, 10 firms, rank 0	Common sorting 0.8	T=7	0, 1
Rank 1, independent	80 workers, 10 firms, singular value 1	Independent	T=7	0, 1
Rank 1, common sorting	80 workers, 10 firms, singular value 1	Common sorting 0.8	T=7	0, 1
Rank 1, interaction sorting	80 workers, 10 firms, singular value 1	Interaction sorting 0.4	T=10	0, 1
Grouped BLM	300 workers, 18 firms; 2 worker types x 3 firm classes	Common 0.4; interaction 0.2	T=5	0, 1
Rank 2	80 workers, 10 firms; singular values 1 and 0.5	Common 0.4; interaction 0.4	T=15	0, 1, 2

All designs use wage-noise standard deviation 0.5. The redraw probability is 0.75 except in the grouped design, where it is 0.35. There are 100 replications per design. The production run contains 32,400 scalar estimate-target records and 5,200 estimator attempts: 4,773 successful, 427 unstable, and zero hard failures.

For replication errors e_b , the reported bias, Monte Carlo standard error, and RMSE are

$$\text{bias} = \text{mean}_b(e_b),$$

$$\text{MCSE}(\text{bias}) = \text{sd}_b(e_b) / \sqrt{B},$$

$$\text{RMSE} = \sqrt{\text{mean}_b(e_b^2)}.$$

The primary result is unconditional and includes finite values returned by unstable attempts. Stable-only results are separately labeled robustness calculations; they do not replace the unconditional procedure result.

Source: configs/full_ladder.json; PRODUCTION_RESULTS.md; src/loo_sim/monte_carlo.py. Formulas are derived from the stored replication-level errors.

4. Results

4.1 Additive designs: common targets

Additive designs: common targets, different finite-sample behavior

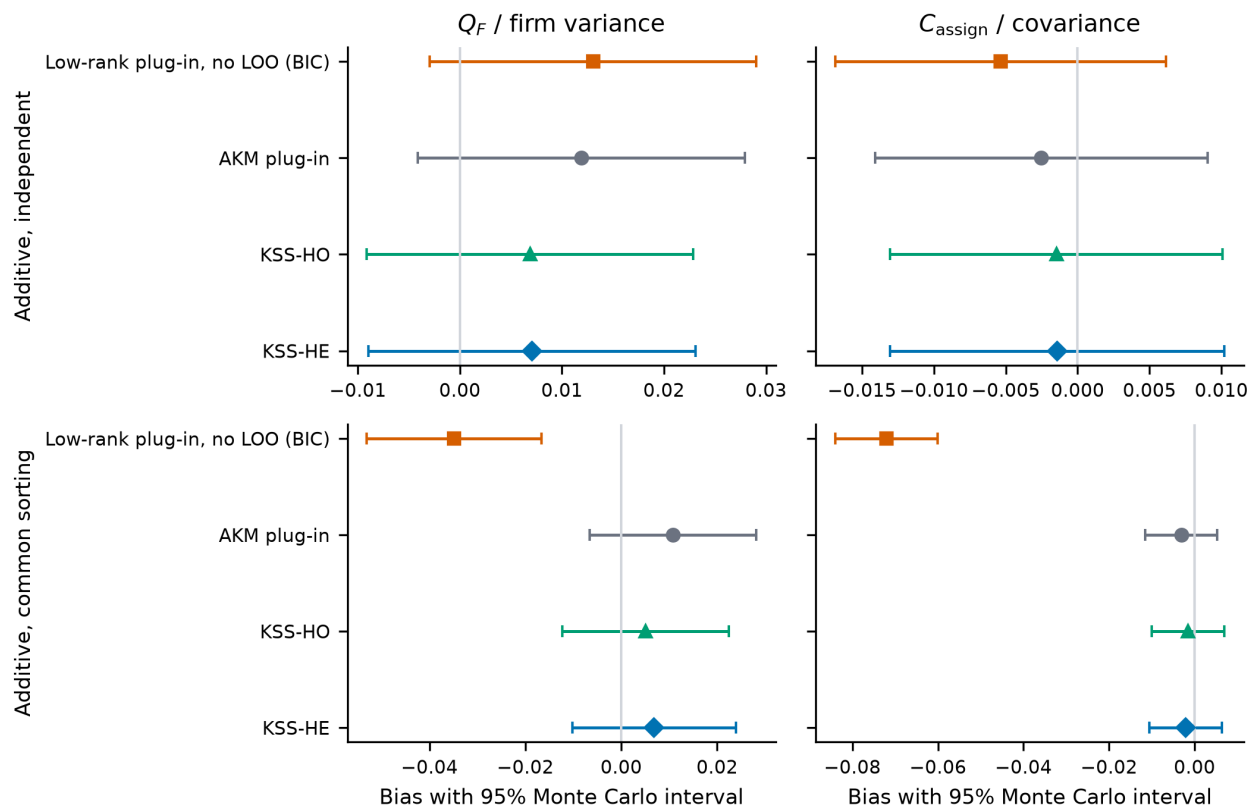


Figure 1. Additive-design bias and 95 percent Monte Carlo intervals.

Under independent assignment, the low-rank plug-in without LOO correction, AKM plug-in, KSS-HO, and KSS-HE have nearly identical RMSE. The low-rank plug-in bias is 0.013 for Q_F and -0.005 for C_{assign} ; KSS-HE bias is 0.007 and -0.001, respectively.

Under common sorting, the targets still coincide, but the low-rank plug-in has bias -0.035 (MCSE 0.009) for Q_F and -0.072 (0.006) for C_{assign} . KSS-HE bias is 0.007 (0.009) and -0.002 (0.004). Thus the sorted-design difference is finite-sample plug-in behavior, not an estimand gap. It is the kind of bias the future project correction is intended to address, but this experiment does not establish what the unfinished correction will do.

Source: reports/production/tables/additive_comparison.csv. Derived interpretation: equality of targets follows from additive separability in Section 1.

4.2 Correct rank is not a stability certificate

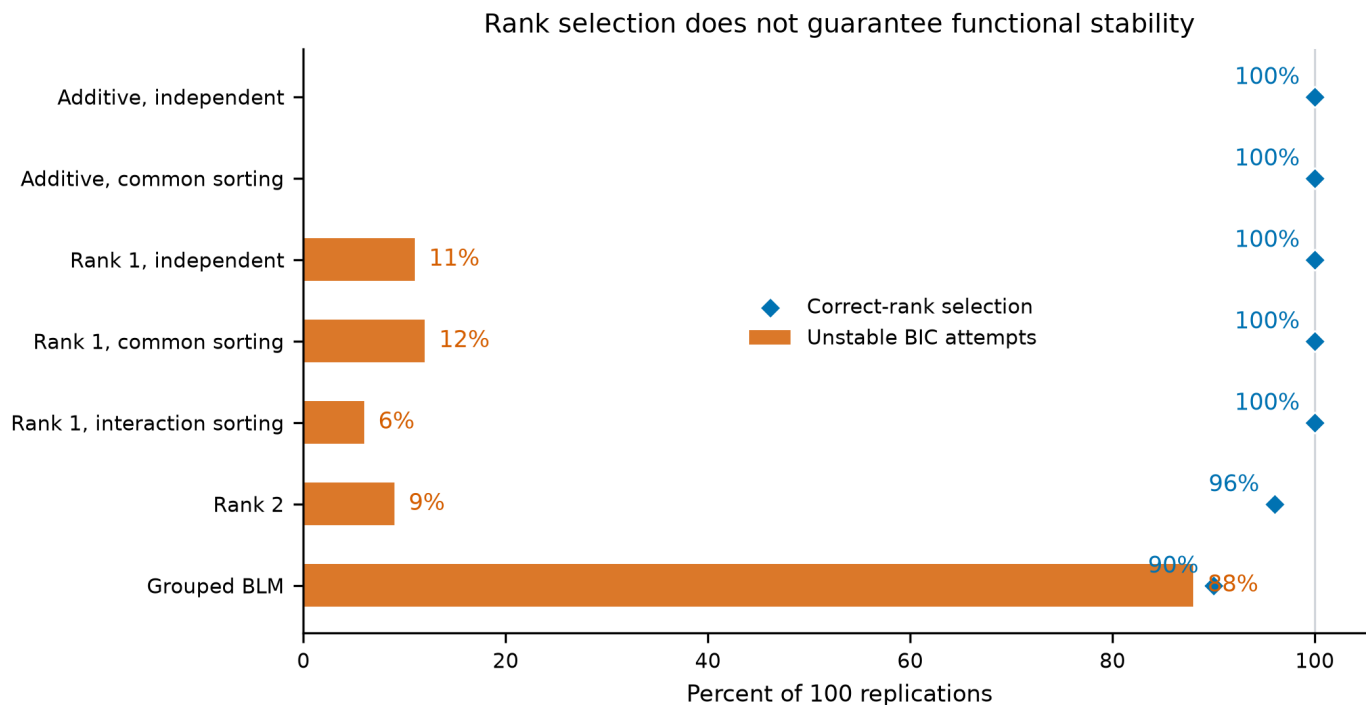


Figure 2. BIC rank selection and unstable-attempt frequencies.

BIC selects the true rank in every additive and rank-one free-factor replication and in 96 percent of rank-two replications. Nevertheless, 6-12 percent of BIC fits are unstable in the rank-one and rank-two free-factor designs. In the grouped design, BIC selects rank one in 90 percent of replications but is unstable in 88 percent.

Derived interpretation. Rank selection answers which candidate fits best on the common observed support. Functional stability additionally requires the completed schedule to produce reliable off-support project functionals. The first property does not imply the second.

Source: reports/production/tables/stability_rank_selection.csv and the stored attempt diagnostics.

4.3 Nonadditivity: stable fits versus unconditional risk

Nonadditive low-rank plug-in (no LOO): instability drives tail risk

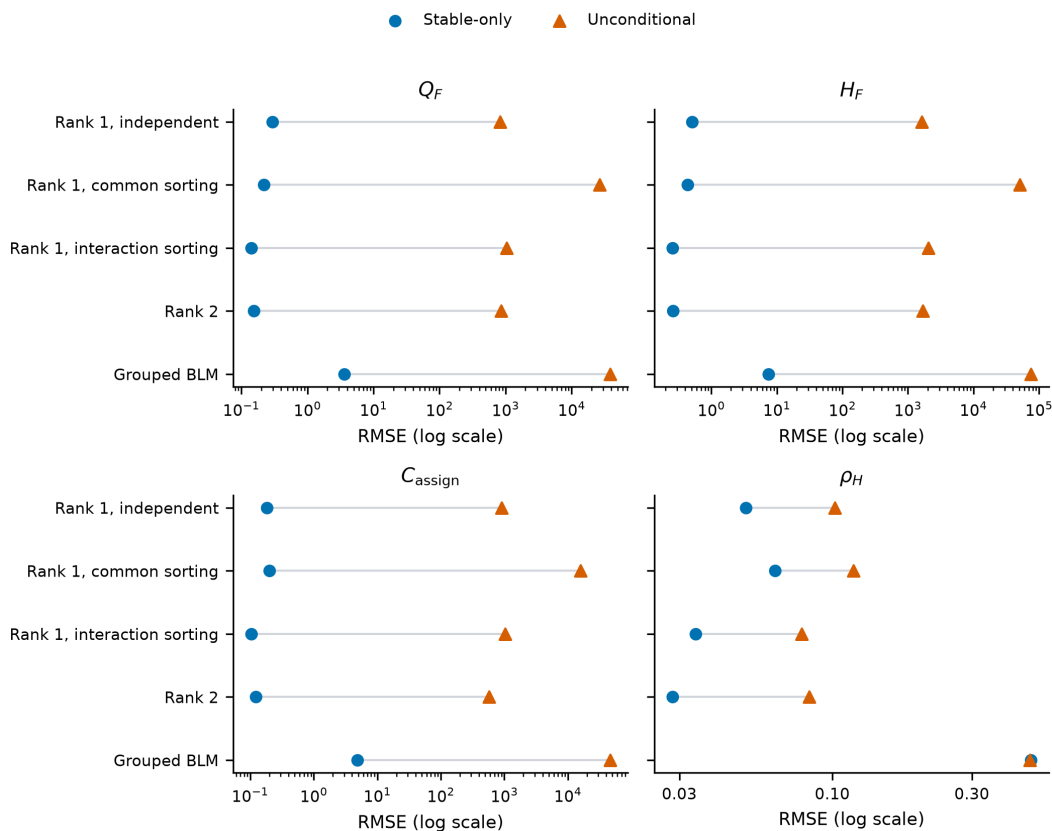


Figure 3. Unconditional and stable-only RMSE for project functionals.

Conditional on a stable fit, free-factor RMSE ranges from 0.142 to 0.296 for Q_F , 0.255 to 0.513 for H_F , 0.104 to 0.201 for C_{assign} , and 0.028 to 0.064 for ρ_H . These values show that stable completion can recover the four schedule-based objects with moderate error.

The unconditional conclusion is different. The 6-12 percent unstable free-factor attempts generate very large functionals, producing unconditional Q_F RMSE between 823 and 26,563. In the grouped design, stable-only Q_F RMSE is already 3.586, only 12 fits are stable, and unconditional Q_F RMSE is about 38,100.

Derived interpretation. The stable-only panel diagnoses what happens after conditioning on a fit-dependent screen. It is informative about the numerical failure mode but cannot be the headline performance measure. The unconditional panel shows the operational risk of using the current plug-in procedure without an additional stabilization or correction step.

Source: reports/production/tables/nonadditive_comparison.csv; PRODUCTION_RESULTS.md, nonadditive-results section.

4.4 KSS can estimate its own target while missing Q_F

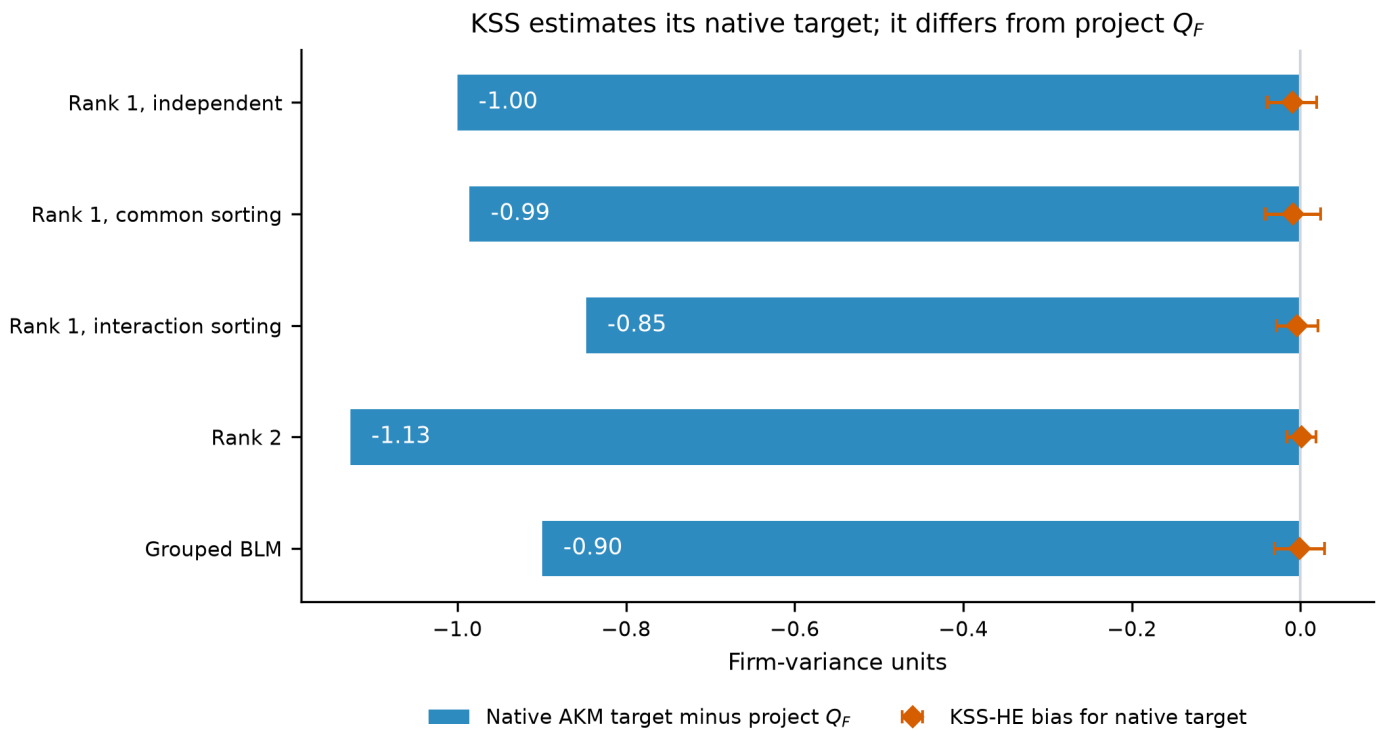


Figure 4. Native AKM/KSS target gaps and KSS-HE native-target bias.

Across the five nonadditive designs, the native AKM firm-variance target is 0.847 to 1.127 below project Q_F . By contrast, KSS-HE bias relative to that native target ranges from -0.010 to 0.001, with MCSE between 0.009 and 0.017.

Derived interpretation. Small native-target estimation error is compatible with a large error relative to the project estimand. This is not a failure of KSS: it is an estimand difference created by projecting a nonadditive schedule onto an additive model. KSS also does not produce project H_F or ρ_H .

Source: reports/production/tables/nonadditive_comparison.csv; native-target definitions in SIMULATION_CONTRACT.md, Section 5.1.

4.5 The grouped design favors grouped BLM

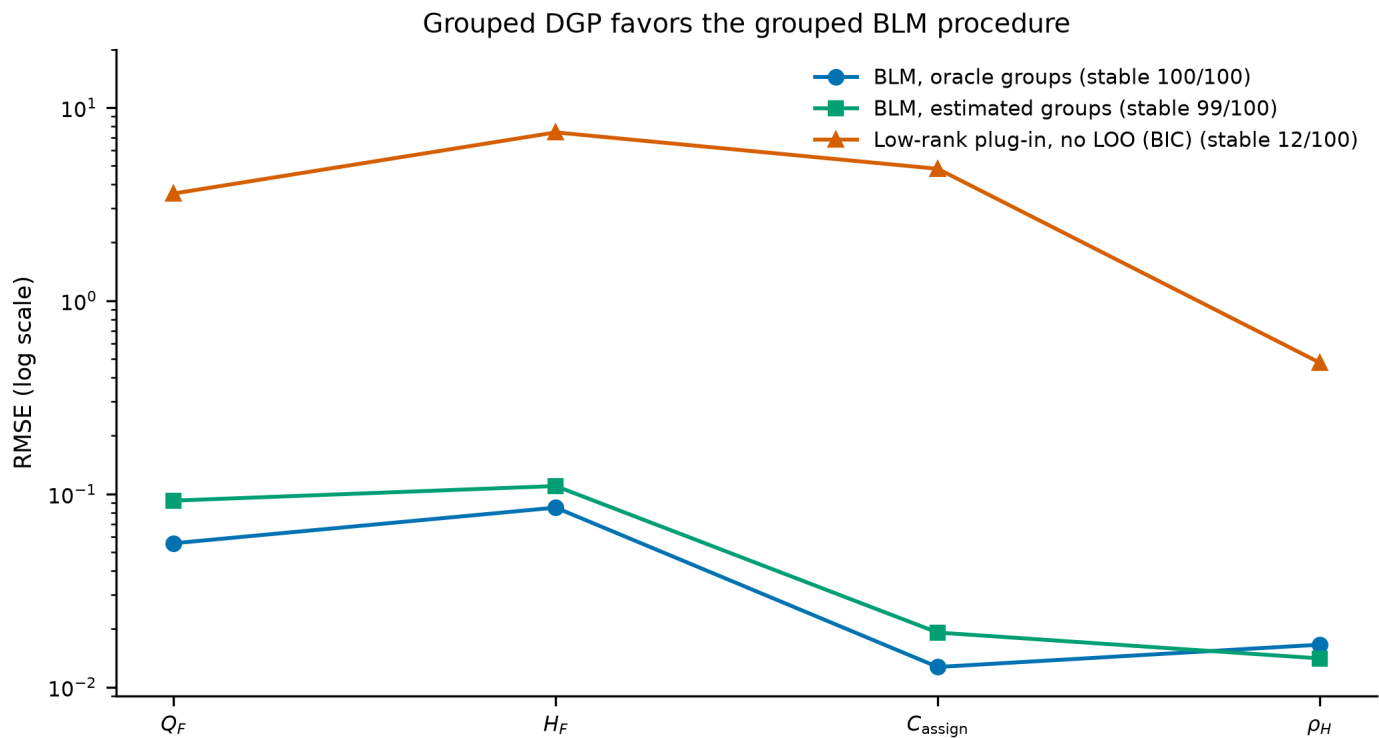


Figure 5. Stable-only RMSE in the grouped BLM design.

Oracle-group BLM is stable in 100 of 100 replications; estimated-group BLM is stable in 99. Their aligned cell-mean RMSE is about 0.051. For the four grouped project functionals, oracle-group RMSE is 0.056, 0.085, 0.013, and 0.017; estimated-group RMSE is 0.092, 0.110, 0.019, and 0.014. The low-rank plug-in is stable in only 12 replications, with stable-only RMSE 3.586, 7.431, 4.814, and 0.478.

Derived interpretation. The grouped BLM procedure uses the correct dimension reduction for this DGP. A low matrix rank does not by itself make individual-level low-rank completion reliable when the observed support is weak and the economically relevant structure is discrete worker-type by firm-class heterogeneity.

Source: reports/production/tables/grouped_blm_comparison.csv. The final sentence is a comparison derived from the DGP and the reported stability and RMSE values.

4.6 Borovičková-Shimer estimates a different sorting object



Figure 6. Bias of the Borovičková-Shimer native worker-firm correlation estimator.

The implemented Borovičková-Shimer procedure returns a value in every replication but is negatively biased for its own native correlation target. Bias ranges from -0.049 in the grouped design to -0.404 in the rank-two design; the additive-independent bias is -0.149. The procedure also uses a smaller valid sample after removing return spells and insufficiently observed workers or firms.

Derived interpretation. These results are finite-sample properties of the implemented cleaning and estimation pipeline, not evidence that $\rho_{BS} = C_{assign}$. The two objects remain conceptually different: one is a correlation between matched worker and firm wage types, while the other is a variance difference for the complete schedule.

Source: reports/production/tables/bs20_native.csv; SIMULATION_CONTRACT.md, Section 3; PyTwoWay 0.3.21 wrapper in src/loo_sim/pytwo_way_estimators.py.

5. Interpretation and limits

The experiment yields three conclusions that should remain separate.

- 1 **Native-estimand performance.** KSS has small average bias for the additive projection in these designs; BLM performs well for the grouped target; Borovičková-Shimer exhibits material negative finite-sample bias for its native correlation.
- 2 **Project-estimand performance.** The low-rank plug-in without LOO correction can recover Q_F , H_F , C_{assign} , and ρ_H when completion is stable, but it has conditional plug-in bias and severe unconditional tail risk.
- 3 **Estimand gaps.** Under nonadditivity, good AKM/KSS estimation does not imply recovery of the schedule-based project objects. The firm-variance gap is economically and numerically large in every calibrated nonadditive design.

The simulation does not evaluate standard errors for the unfinished project estimator, prove that its eventual LOO correction removes the observed bias, or establish that the present stability diagnostic is optimal. The next controlled experiment is to freeze this DGP ladder and replace the current plug-in with the completed leave-worker-out estimator once its estimand, correction, and feasible standard-error formula are finalized.

Derived synthesis from Figures 1-6 and their source tables. Scope limitation: SIMULATION_CONTRACT.md, estimator-label and reporting contracts.

References

- Bonhomme, Stéphane, Thibaut Lamadon, and Elena Manresa. 2019. "A Distributional Framework for Matched Employer Employee Data." *Econometrica* 87(3): 699-739. <https://doi.org/10.3982/ECTA15722>.
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- PyTwoWay 0.3.21. Python implementation used for the FE/KSS, BLM, and Borovičková-Shimer comparison procedures. <https://github.com/tlamadon/pytwoway>.
- Project simulation sources: `SIMULATION_CONTRACT.md`, `configs/full_ladder.json`, `PRODUCTION_RESULTS.md`, and `reports/production/report_metadata.json`.